

1) What is the MS-part of ANOVA-table?

Where does it come from?

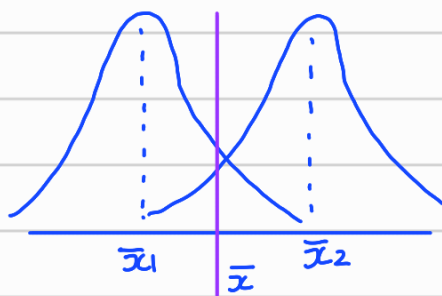
The MEAN SQUARES or MEAN SUM OF SQUARES column contains the AVERAGE sum of squares: SS/df .

Mean squares are used to determine whether treatments are significant

2) Why do we do tests based on variation if we want to conclude something about the equality of group means?

Due to sampling error, we expect that drawing different samples from the same distribution will lead to different means being observed within groups, but does this mean there is a TRUE, SIGNIFICANT difference between means?

Between group variation

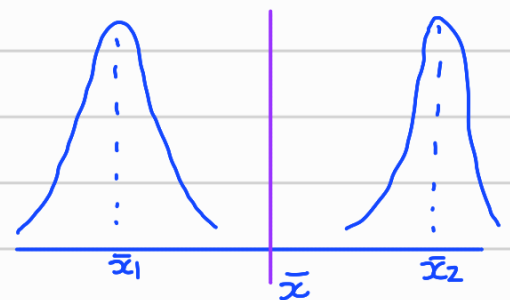


Samples overlap: expect individual means to be quite similar / not differ by a great margin

ALSO

difference between individual and overall means not expected to be significant

VS



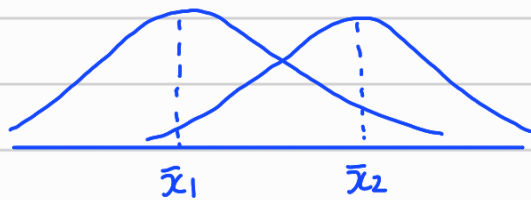
samples differ by great margin: expect individual means to differ

ALSO

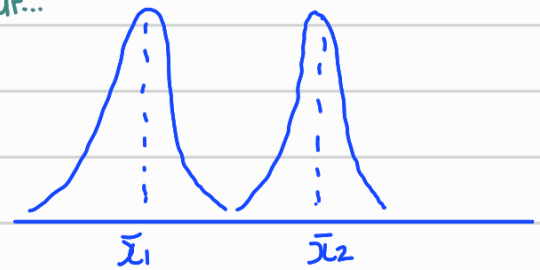
difference between individual means and overall mean would likely be significant

Within-group variation

similar means but...



VS



As spread of each sample increases, distributions overlap (all part of one big population)

less variability in each sample: seem to belong to different populations

We evaluate **each sample individually**: how far is each observation from its own group mean?



- 3) Is $\bar{y} = \frac{\bar{y}_1 + \bar{y}_2 + \dots + \bar{y}_n}{n} = \frac{\sum_{i=1}^m y_i}{m}$ where $n = \#$ of groups and $m = \#$ of observations? **YES!**
- $\bar{y}$ is the overall mean: the mean of sample means OR the mean of all observations combined, as long as each group has the same $\#$ of observations e.g. **5** students in each of 3 tutor groups.
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